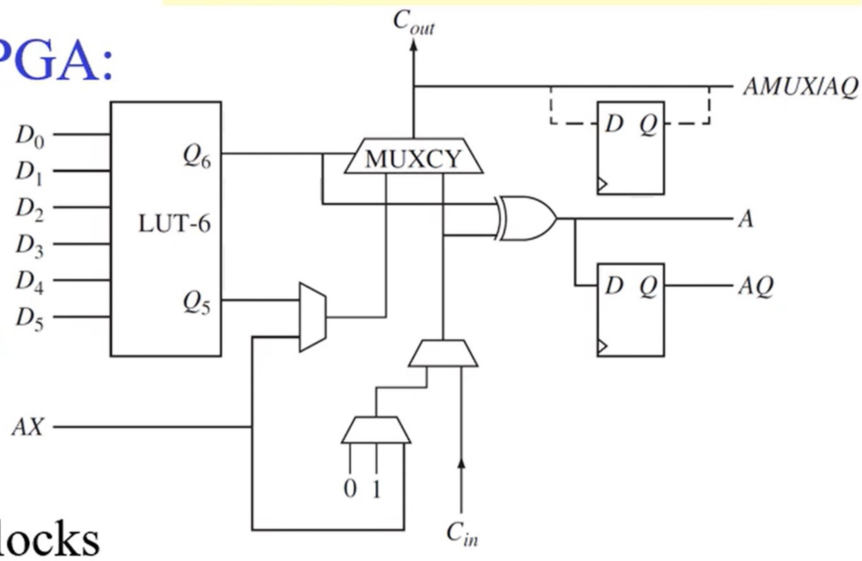
6.5 Examples of Logic Blocks in Xilinx FPGAs

Xilinx Kintex

* The LUT6 can be used to generate one 6-variable function (Q6) or two 5-variable functions (Q5,Q6).

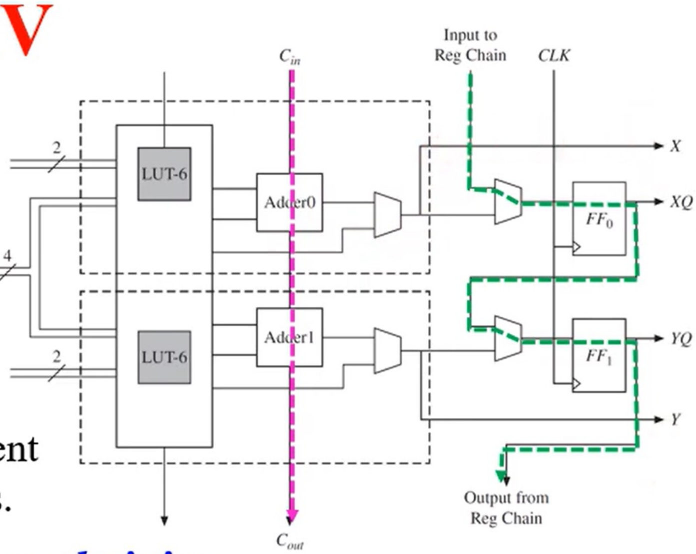
- Xilinx Kintex FPGA:
  - ***Basic block***:
    - a LUT6
    - 2 flip-flops
    - the ***carry chain***
    - several MUXs



  - ***Slice***: 4 basic blocks
  - ***Configurable logic block*** (***CL***): 2 slices

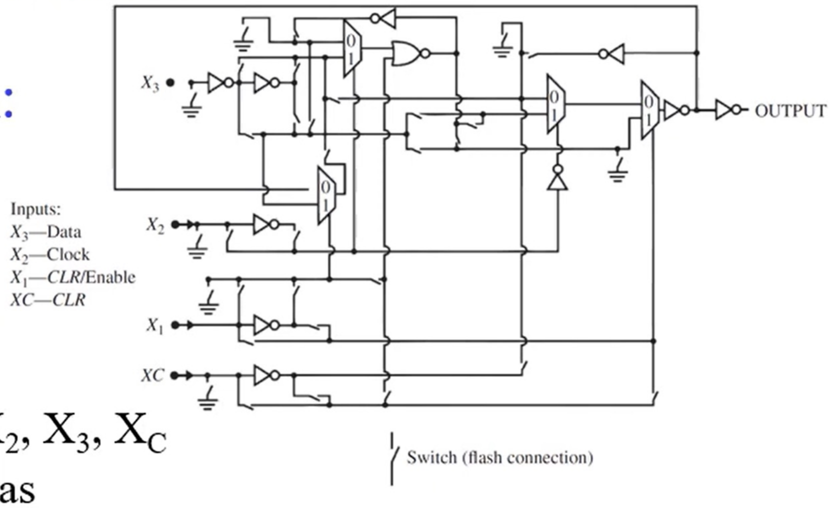
Altera Stratix IV

- Altera Stratix IV FPGA:
  - ***Logic module*** (***LM***): the basic building block
    - Two LUT6
      - each contains 2 independent inputs and 4 shared inputs.
    - Two 1-bit adders with ***carry-chaining***
    - Two flip-flops with ***register chaining***
      - allows creation of ***shift registers*** or other types of ***flip-flop arrays*** and use the flip-flops separately from LUTs.



Microsemi Fusion

- Microsemi Fusion:
  - *VersaTile*:
    the building block
    - consists of MUXs and gates.
    - has 4 inputs, X_1, X_2, X_3, X_C
    - can be configured as
      a 3-input logic function (as $F(X_1, X_2, X_3)$),
      a latch with a clear or set,
      a D flip-flop with clear or set, or
      a D flip-flop with enable, clear, or set.



  - is of ***finer granularity*** compared to an LUT4